

Agent Banking Fraud Detection — Oracle Machine Learning (OML) Design

Source Table: `EMOB.MB_TRANSACTION_DTL` | Target: Oracle Database 19c

1. Fraud Detection Use Cases (30+ Scenarios)

Velocity & Structuring

1. **Structuring/Smurfing** — multiple deposits just under reporting thresholds (e.g., BDT 99,000 repeated instead of one BDT 500,000+ transaction).
2. **Rapid deposit-then-withdrawal** — cash-in followed by cash-out within minutes, same or linked account, indicating pass-through laundering.
3. **Velocity fraud** — abnormal spike in transaction count per agent/customer within a short rolling window (e.g., >10 transactions/hour vs. historical baseline of 2/hour).
4. **Burst micro-transactions** — many small-value transactions in rapid succession to avoid single-transaction monitoring.
5. **Round-trip/circular transactions** — funds moving $A \rightarrow B \rightarrow C \rightarrow A$ within a short time window, no genuine economic purpose.

Behavioral & Account Anomalies

6. **Dormant account reactivation abuse** — account inactive for 90+ days suddenly receiving/sending large sums.
7. **Sudden behavior deviation** — customer's transaction size/frequency deviates sharply (z-score) from their own historical pattern.
8. **New account high-value activity** — account opened <7 days ago already transacting at high value/frequency.
9. **Account takeover pattern** — sudden change in device, location, and transaction behavior simultaneously.
10. **Beneficiary concentration** — one CR_AC_NO receiving funds from an unusually large number of distinct DR_AC_NO in a short period (fan-in / money mule pattern).

Agent-Side Fraud

11. **Fake commission generation** — commission credits not proportionate to genuine transaction volume.
12. **Agent self-dealing** — agent's own account frequently appearing as counterparty in their processed transactions.

13. **Agent collusion ring** — cluster of agents repeatedly transacting with the same small set of customer accounts.
14. **Ghost transaction reversal abuse** — high rate of reversed/failed transactions followed by successful re-attempts at different amounts.
15. **Agent point cash imbalance** — TRAN_POINT_ID cash-out volume inconsistent with reported float/liquidity.
16. **Off-hours agent activity** — agent processing transactions well outside registered business hours.
17. **Agent commission spike without matching transaction growth** — commission credited disproportionately to actual TRAN_AMT_LC volume.

Device & Technical Anomalies

18. **Device sharing across agents** — same DEVICE_ID used to log in/process for multiple distinct CREATE_BY (agent) IDs.
19. **Device sharing across customers** — same DEVICE_ID initiating transactions for many unrelated AC_ID/customer accounts.
20. **New/unrecognized device high-value transaction** — first-seen DEVICE_ID for a customer immediately used for a large transaction.
21. **SIM/device swap followed by transaction spike** — device fingerprint change immediately preceding unusual activity.
22. **Duplicate transaction attempts** — same DOC_NUM/REF_DOC_NO or matching amount+account+timestamp cluster suggesting replay or double-submission fraud.

Geographic Anomalies

23. **Impossible travel** — LAT/LNG for consecutive transactions imply a travel speed physically impossible (e.g., 300 km in 10 minutes).
24. **Geographic hotspot clustering** — unusual concentration of high-risk transactions from a specific LAT/LNG cluster inconsistent with the agent point's registered location.
25. **Cross-border-adjacent proximity anomaly** — transactions clustering suspiciously near border regions with irregular timing (potential trafficking-linked flows).
26. **Branch-agent location mismatch** — TRAN_POINT_ID's registered location differs significantly from LAT/LNG of the actual transaction.

Amount & Statistical Anomalies

27. **High-value abnormal transaction** — TRAN_AMT_LC far above the customer's or agent point's historical percentile (e.g., >99th percentile).

28. **Threshold-hugging pattern** — repeated transactions clustering just below known KYC/AML reporting thresholds.
29. **Unusual DR/CR imbalance** — cash-in-to-cash-out ratio for an agent point deviates sharply from peer agents in the same region.
30. **Round-number bias** — disproportionate frequency of suspiciously "clean" round amounts (often indicative of manual/fabricated entries vs. organic customer behavior).

Additional Innovative Scenarios

31. **Synthetic identity clustering** — multiple AC_ID sharing identical or near-identical KYC metadata but transacting independently (evasion of per-customer limits).
 32. **Commission-transaction time lag anomaly** — abnormal delay/advance between the underlying transaction and its commission posting, suggesting manual manipulation.
 33. **Inter-branch transaction laundering chain** — funds routed through 3+ branches in a short window with no plausible business reason.
 34. **CBS_STATUS manipulation pattern** — unusually high rate of status overrides/manual corrections associated with a specific agent or branch.
 35. **Seasonal/festival exploitation** — fraud rings timing activity around Eid/payroll cycles when monitoring thresholds are typically relaxed for legitimate remittance spikes.
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2. Fraud Features for the OML Model

Core Transactional Features

- `TRAN_AMT_LC` (raw and log-transformed)
- Transaction hour, day-of-week, is-weekend, is-holiday flag
- `TRAN_TYPE`, `TRAN_CODE`, `DRCR_CODE` (categorical, target/frequency encoded)
- Time since account's last transaction (recency)
- Time since agent's/device's last transaction

Behavioral Aggregates (rolling windows: 1hr / 24hr / 7d / 30d)

- Transaction count per AC_ID, per CREATE_BY (agent), per DEVICE_ID
- Sum/average/std-dev of `TRAN_AMT_LC` per entity per window
- Cash-in vs. cash-out ratio per agent point
- Number of distinct counterparties (DR_AC_NO / CR_AC_NO) per entity
- Number of distinct devices used per customer; number of distinct agents per device

Geographic Features

- Distance (haversine) between consecutive transaction LAT/LNG for same entity
- Implied velocity (distance/time) between consecutive transactions
- Distance from agent point's registered/home coordinates
- Geographic entropy (spread of transaction locations) per customer/agent

Network/Relational Features

- Degree centrality of AC_ID in the counterparty transaction graph (fan-in/fan-out count)
- Shared-device graph: number of unique customers/agents linked via common

DEVICE_ID
- Cycle detection flag (A→B→C→A within N hours)

Engineered / Derived Features

- Z-score of current transaction amount vs. entity's own historical mean/std-dev
- Benford's Law digit-distribution deviation score per agent (amount-manipulation signal)
- New-device flag, new-counterparty flag, new-branch flag (binary "first-seen" indicators)
- Threshold-proximity score (distance from known regulatory reporting thresholds)
- Commission-to-transaction ratio vs. peer-agent baseline
- Reversal/failure rate per agent over trailing 30 days

3. Machine Learning Approach — Algorithm Comparison

Approach	OML Algorithm	Best Use Case	Notes
Classification (supervised)	GLM (Logistic Regression), Random Forest, SVM, Naïve Bayes, Neural Network	When labeled fraud/non-fraud history exists (e.g., confirmed AML cases, chargebacks)	Highest precision when labels are reliable; needs a labeled dataset which is often scarce early on
Anomaly Detection (unsupervised)	One-Class SVM, Expectation	Detecting novel fraud patterns with no prior labels — the realistic	Learns "normal" behavior and flags

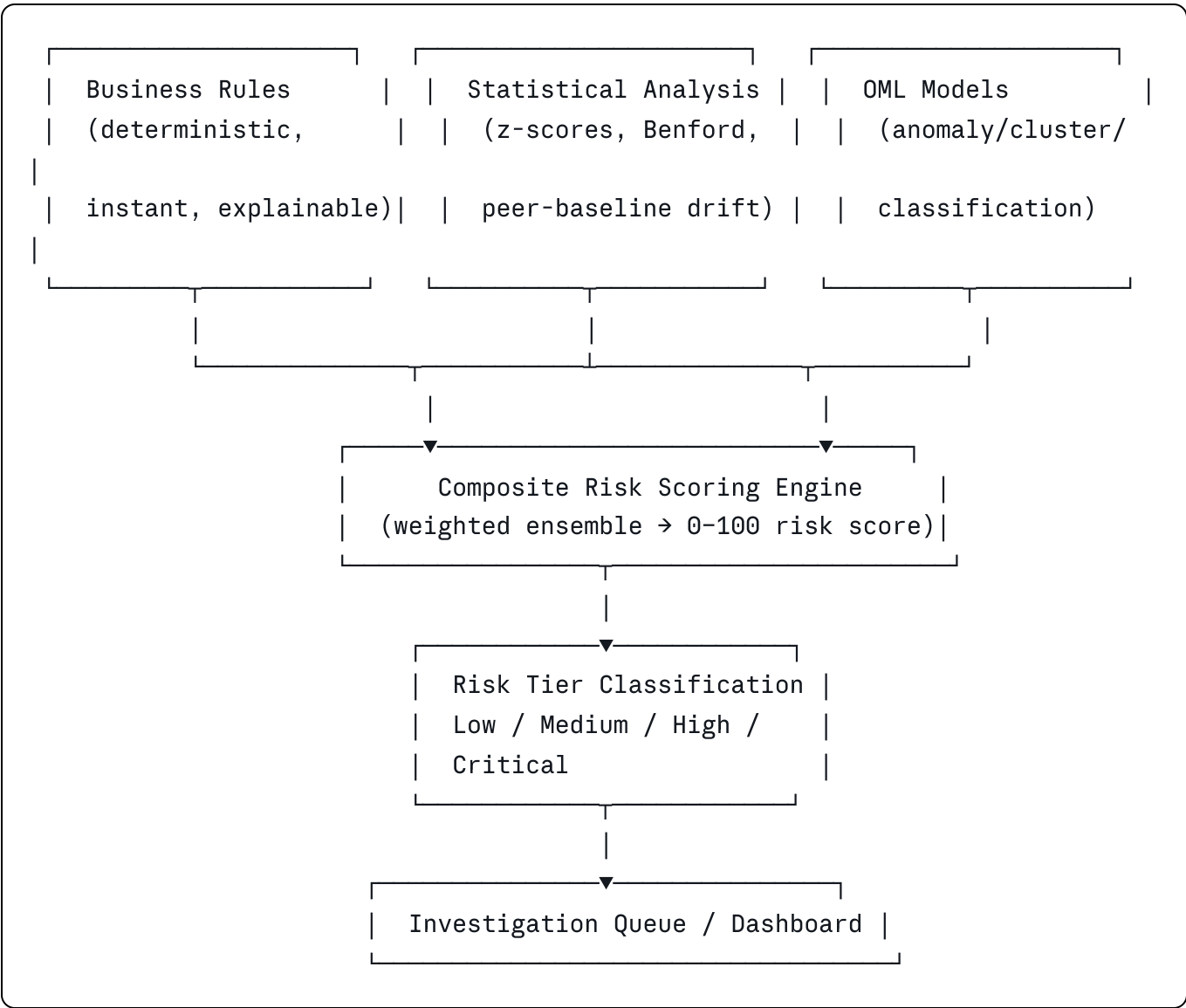
Approach	OML Algorithm	Best Use Case	Notes
Clustering	Maximization (Clustering-based)	starting point for most Agent Banking systems	deviation; doesn't need fraud labels
	K-Means, O-Cluster, Expectation Maximization	Peer-grouping agents/customers to build behavioral baselines and detect outlier clusters	Useful for segmenting agent points by risk profile before scoring
Decision Trees	Decision Tree (CART)	Rule-extraction and explainability — converting ML insight into audit-friendly business rules	Excellent for regulator- facing explainability
Random Forest	Random Forest	Robust general-purpose classifier once labeled fraud data accumulates	Handles nonlinear interactions between features well, resistant to overfitting
SVM	Support Vector Machine (Linear/Gaussian)	High-dimensional feature spaces (many engineered features), strong for one- class fraud boundary modeling	Computationally heavier; needs feature scaling
GLM	Generalized Linear Model	Baseline interpretable risk-scoring model; good first production model due to transparency	Coefficients map directly to explainable risk factors
Naïve Bayes	Naïve Bayes	Fast, lightweight first- pass filter, especially for categorical fraud indicators (device/agent flags)	Assumes feature independence — best as a pre-filter, not final scorer
Neural Networks	OML Neural Network (Oracle 19c supports basic feed-forward NN)	Capturing complex nonlinear behavioral patterns once sufficient volume/history exists	Least explainable; best combined with an explainability layer (Section 9)

Recommended Progression

- 1. **Phase 1 (no labels yet):** Unsupervised anomaly detection (One-Class SVM) + clustering (K-Means/O-Cluster) for peer-group baselining.
- 2. **Phase 2 (rules generate initial labels):** Use business-rule flags as weak labels to train a Decision Tree / Random Forest classifier.
- 3. **Phase 3 (confirmed investigation outcomes accumulate):** Retrain GLM/Random Forest/Neural Network on confirmed fraud/non-fraud outcomes for supervised scoring.

4. Hybrid Fraud Detection Strategy

A single technique is never sufficient in Agent Banking — rules catch known patterns instantly but miss novel fraud; ML catches novel patterns but needs time/data and can be a black box to auditors. The recommended architecture layers all four:



How they complement each other:

- **Rules** provide zero-latency, regulator-mandated hard stops (e.g., threshold-hugging, blacklisted accounts) — auditable and immediately actionable.
- **Statistical analysis** (per-entity z-scores, Benford's Law, peer drift) catches subtler deviations rules can't hardcode, and is still fully explainable.
- **OML models** catch complex, multivariate, and evolving fraud patterns that neither static rules nor simple statistics can express — particularly collusion rings and behavioral drift.
- **Composite scoring** combines all three into one number analysts can act on, rather than three disconnected alert streams causing alert fatigue.

5. Fraud Risk Levels & Scoring

Risk Tier	Score Range	Characteristics	Recommended Action
Low	0-24	Normal behavioral pattern, no rule triggers, ML anomaly score near baseline	Log only, no analyst action
Medium	25-49	One soft rule trigger OR moderate statistical deviation (e.g., z-score 2-3) OR mild ML anomaly score	Auto-flag for periodic batch review, no transaction hold
High	50-74	Multiple rule triggers OR strong ML anomaly score OR confirmed geographic/device anomaly	Route to analyst queue within SLA (e.g., 4 hours); optional soft transaction hold pending review
Critical	75-100	Hard-rule breach (e.g., structuring, impossible travel, known blacklist match) combined with high ML confidence	Immediate hold/block, real-time alert to AML/compliance officer, mandatory investigation

Scoring weights (e.g., 40% rules, 25% statistical, 35% ML) should be tunable and periodically recalibrated against confirmed investigation outcomes.

6. Key Fraud Indicators

- **Behavioral anomalies:** deviation from entity's own historical transaction pattern (amount, frequency, timing)
 - **Geographic anomalies:** impossible travel, location-agent mismatch, hotspot clustering
 - **Device anomalies:** device sharing, new-device-high-value, device/SIM swap correlation
 - **Velocity anomalies:** transaction count/amount spikes within short rolling windows
 - **Amount anomalies:** threshold-hugging, round-number bias, statistical outliers (z-score/Benford deviation)
 - **Account relationship anomalies:** fan-in/fan-out counterparty concentration, circular transaction chains
 - **Agent activity anomalies:** commission-to-volume mismatch, off-hour activity, high reversal rates
 - **Customer activity anomalies:** dormant reactivation, sudden new-counterparty surge, KYC-metadata overlap with other accounts (synthetic identity signal)
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7. Dashboards & Investigation Tooling

1. **Real-Time Fraud Monitoring Dashboard** — live feed of Critical/High risk transactions with drill-down.
2. **Suspicious Transaction Review Queue** — analyst worklist with status tracking (New / In Review / Escalated / Closed).
3. **Agent Risk Ranking Dashboard** — leaderboard of agents by composite risk score, commission anomaly, reversal rate.
4. **Customer Risk Ranking Dashboard** — customers ranked by cumulative risk score and flagged-transaction count.
5. **Device Risk Analysis View** — device-to-agent and device-to-customer sharing graphs, flagged device list.
6. **Geographic Fraud Hotspot Map** — LAT/LNG heatmap of flagged transactions overlaid on agent point locations.
7. **Daily Fraud Alert Digest** — summarized email/report of new Critical/High alerts for compliance officers.
8. **Executive Fraud Summary** — trend charts (fraud rate over time, financial exposure, tier distribution, model precision/recall trend) for management reporting.
9. **Case Investigation Workbench** — full transaction history, linked-entity graph, and score breakdown for a flagged case, supporting SAR/STR filing workflow.

8. Model Evaluation Metrics

Metric	Why It Matters in Fraud Detection
Precision	Minimizes false alarms that waste analyst time and erode trust in the system — critical given limited AML staff capacity
Recall	Measures how many actual fraud cases are caught — missing fraud has direct financial and regulatory consequences
F1-Score	Balances precision/recall when class imbalance (fraud is rare) makes accuracy misleading
ROC-AUC	Evaluates the model's ability to rank fraud higher than non-fraud across all thresholds — useful for comparing model versions
False Positive Rate	Directly measures analyst workload/alert fatigue — too high erodes operational trust in the system
False Negative Rate	Directly measures undetected fraud exposure — the costliest error type in AML/fraud contexts
Precision-Recall AUC	More informative than ROC-AUC under severe class imbalance (fraud is typically <1% of transactions)

Because fraud is a rare-event problem, accuracy alone is misleading (a model predicting "no fraud" always could show >99% accuracy); **PR-AUC and Recall at a fixed False Positive budget** should be the primary tuning targets.

9. Explainable AI (XAI) Layer

- **Top contributing risk factors per flagged transaction** — e.g., "Amount 4.2 std-dev above customer baseline; new device; 380km implied travel speed."
- **Feature importance** — global (which features matter most across the model) and local (SHAP-style per-transaction contribution) reporting.
- **Human-readable explanation templates** — auto-generated plain-language summaries for each alert (e.g., "This transaction was flagged primarily due to unusual velocity and a first-seen device").

- **Confidence/risk score with tier label** — always pair the numeric score with the tier (Low/Medium/High/Critical) and top 3 driving factors so analysts don't have to interpret raw model output.
 - **Rule-vs-ML attribution split** — clearly show whether an alert originated from a deterministic rule, statistical deviation, or ML anomaly score, since each has different audit/regulatory implications.
 - **Decision Tree "shadow model"** — optionally run a simple decision tree alongside black-box models purely for explainability/audit purposes, even if it's not the production scorer.
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10. Innovative Concepts Beyond Standard Practice

1. **Graph-based collusion detection** — build a transaction graph (agents/customers/devices as nodes) and use community detection to surface collusion rings invisible to per-transaction scoring.
2. **Benford's Law agent auditing** — apply digit-distribution analysis to each agent's transaction amounts to flag manually fabricated/manipulated entries.
3. **Behavioral biometrics via device/session metadata** — typing/tap timing patterns (if capturable from the app layer) to detect account takeover beyond simple device-ID checks.
4. **Federated peer-group baselining** — dynamically re-cluster agent points by regional/economic profile so "normal" is judged against true peers, not a single national baseline.
5. **Synthetic identity graph matching** — fuzzy-match KYC metadata (name, NID fragments, phone patterns) across accounts to catch identity-splitting fraud.
6. **Adaptive threshold learning** — let thresholds (not just weights) self-adjust per region/season using rolling statistical baselines instead of static configured limits.
7. **Reverse-money-flow simulation** — for any flagged high-risk account, simulate/trace the full inbound-outbound fund path automatically to pre-build the investigation case file.
8. **Commission-fraud specific model** — a dedicated sub-model scoring agent commission legitimacy independent of the main transaction fraud model, since commission fraud has a distinct signature.
9. **Festival/payroll-cycle-aware modeling** — explicitly model known legitimate seasonal spikes (Eid remittances, garment-sector payroll days) as a feature so the model doesn't over-flag genuine surges.
10. **Explainability-driven rule mining from ML** — periodically extract decision-tree rules from the trained ML model's decision boundaries and propose them as new

candidate business rules, closing the loop between ML discovery and the deterministic rule engine.

11. **Cross-channel correlation** — if other EMOB channels (mobile app, USSD) share customer IDs, correlate activity across channels for a unified customer risk view rather than scoring Agent Banking in isolation.

Suggested Implementation Roadmap

Phase	Focus	OML Components
1. Foundation	Feature engineering views on <code>MB_TRANSACTION_DTL</code> , rule engine, peer-group clustering	K-Means/O-Cluster, statistical baseline views
2. Unsupervised Detection	Deploy anomaly scoring, geographic/device graph checks	One-Class SVM, graph queries
3. Weak-Label Bootstrapping	Use rule/analyst outcomes as training labels	Decision Tree, Random Forest
4. Supervised Refinement	Retrain on confirmed investigation outcomes	GLM, Random Forest, Neural Network
5. Explainability & Ops	XAI layer, dashboards, feedback loop into rule mining	Model explainability views, dashboard integration

Prepared for the EMOB Agent Banking fraud detection initiative. This document covers model design and fraud intelligence strategy only, per your request — no implementation code included. Happy to produce the OML `DBMS_DATA_MINING` build scripts, feature-engineering SQL views, or dashboard mockups as a next step whenever you're ready.